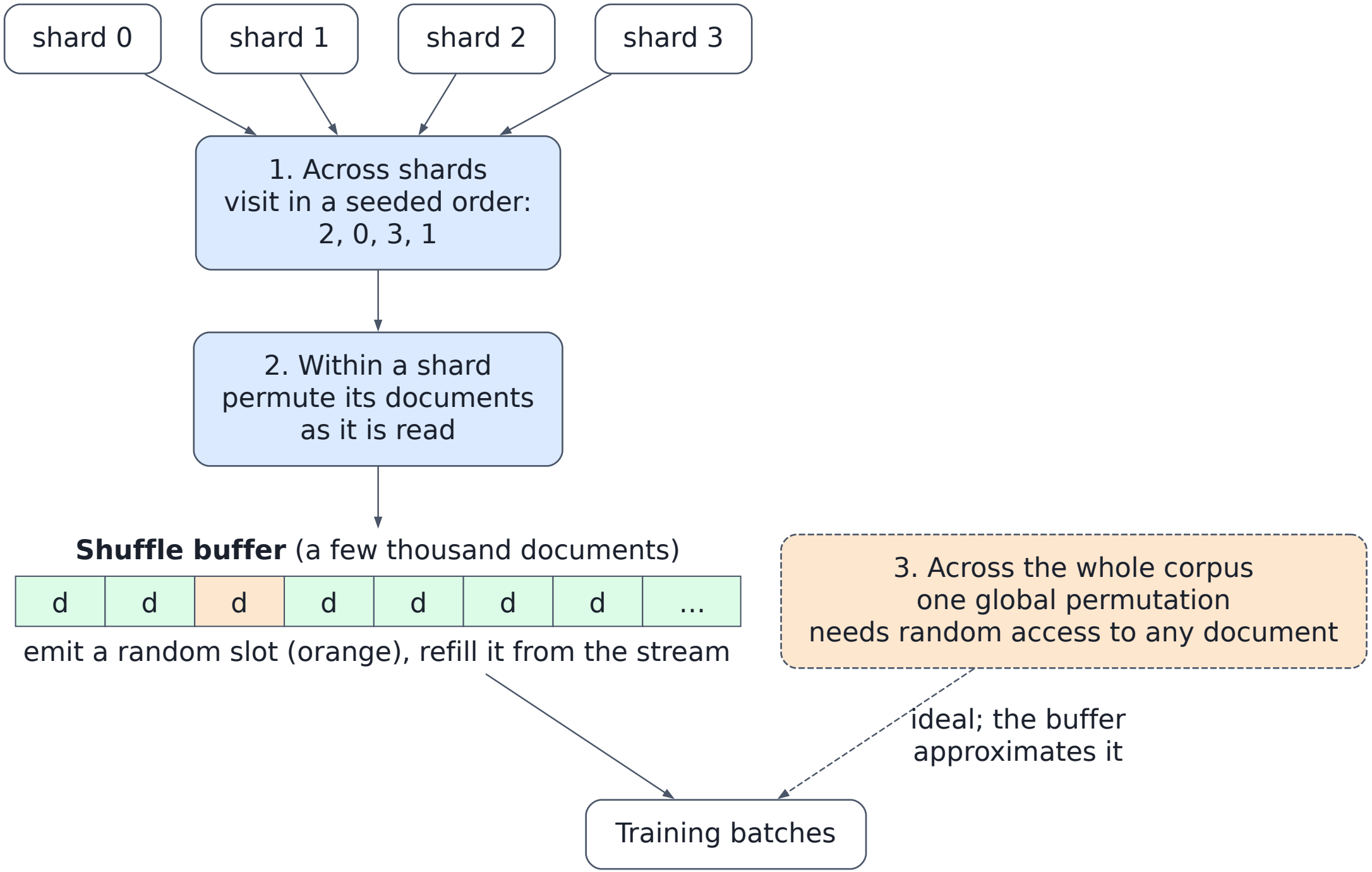
shard 0
shard 1
shard 2
shard 3
1. Across shards
visit in a seeded order:
2, 0, 3, 1
2. Within a shard
permute its documents
as it is read
Shuffle buffer (a few thousand documents)
d
d
d
d
d
d
d
…
emit a random slot (orange), refill it from the stream
3. Across the whole corpus
one global permutation
needs random access to any document
ideal; the buffer
approximates it
Training batches